Our results suggest that the behavior of intermediate features of practical neural networks changes dramatically with increasing overparameterization, in ways that are not obvious from previous analysis, and cannot be easily characterized by properties at initialization.

3 Experimental Setup and Background

Measuring Representation Similarity with CKA: Centered kernel alignment (CKA) [6, 19] addresses several challenges in measuring similarity between neural network hidden representations including (i) their large size; (ii) neuronal alignment between different layers; and (iii) features being distributed across multiple neurons in a layer. Like Nguyen et al. [29], we use the minibatch implementation of linear CKA with a batch size of 256 sampled without replacement from the test dataset, and accumulate statistics over 10 epochs to allow the minibatch estimator to converge.

More concretely, given k minibatches of n examples, and two layers having p_1 neurons and p_2 neurons each, minibatch CKA takes as inputs k pairs of centered activation matrices $(\boldsymbol{X}_1, \boldsymbol{Y}_1), \ldots, (\boldsymbol{X}_k, \boldsymbol{Y}_k)$ where $\boldsymbol{X}_i \in \mathbb{R}^{n \times p_1}$ and $\boldsymbol{Y}_i \in \mathbb{R}^{n \times p_2}$ reflect the activations of these layers to the same minibatches. It produces a scalar similarity score between 0 and 1 by averaging the scores of Hilbert-Schmidt independence criterion (HSIC), computed with a linear kernel, over the minibatches:

$$\mathrm{CKA}_{\mathrm{minibatch}} = \frac{\frac{1}{k}\sum_{i=1}^{k} \mathrm{HSIC}_1(\boldsymbol{X}_i\boldsymbol{X}_i^\top, \boldsymbol{Y}_i\boldsymbol{Y}_i^\top)}{\sqrt{\frac{1}{k}\sum_{i=1}^{k} \mathrm{HSIC}_1(\boldsymbol{X}_i\boldsymbol{X}_i^\top, \boldsymbol{X}_i\boldsymbol{X}_i^\top)}\sqrt{\frac{1}{k}\sum_{i=1}^{k} \mathrm{HSIC}_1(\boldsymbol{Y}_i\boldsymbol{Y}_i^\top, \boldsymbol{Y}_i\boldsymbol{Y}_i^\top)}}, \tag{1}$$

where HSIC_1 is the unbiased estimator of HSIC from Song et al. [34]. This estimator of linear CKA converges to the same value regardless of the batch size.

We compute CKA between *all* layer representations, including before and after batch normalization, activations, and residual connections. In experiments that involve tracking how a model's internal properties (principal components of activations, representation similarity, etc.) change across epochs, we set batch normalization layers to be in training mode, to reduce the difficulty of adapting to batch statistics of the test set when the model has not converged.

The Block Structure Phenomenon: Nguyen et al. [29] use linear CKA to compute the representation similarity for all pairs of layers within the same model and visualizes the result as a heatmap (with x and y axes indexing the layers from input to output). They find a contiguous range of hidden layers with very high representation similarity (yellow squares on heatmaps in Figure 1a) in very deep or wide models, and call this phenomenon the block structure. The block structure arises in networks that are large *relative* to the size of the training set — although small networks do not exhibit block structure when trained on CIFAR-10, they do exhibit block structure on smaller datasets.

Representations of the layers making up the block structure exhibit different representational geometry than the rest of the layers. For layers inside the block structure, the first principal component explains a large fraction of the variance in representations; this is not the case for the other layers or for networks without the block structure [29]. We replicate this observation in Figure 2a. The similarity between layers inside the block structure reflects the alignment of their first principal components, as can be seen from the following decomposition of linear CKA for centered activation

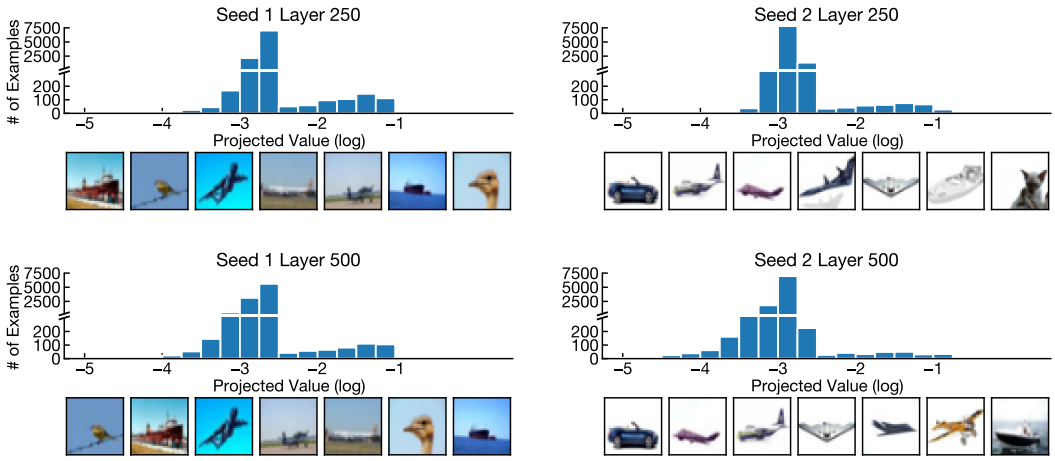

Figure 3: Visualization of the distribution of projected values onto the first principal component by test inputs. There exist a small number of datapoints that yield significantly larger projected values than the rest and dominate the first principal component of the network. Here we show examples of those dominant images for two different seeds of ResNet-164 (1×) (columns). Each seed's dominant images share similar background colors, but these background colors differ between seeds. Further visualization of dominant datapoints across different layers — for instance, layers 250 and 500 in this case — show that they are consistent across layers inside the block structure. See Appendix D for analysis of other models and tasks.

matrices $\boldsymbol{X} \in \mathbb{R}^{n \times p_1}$, $\boldsymbol{Y} \in \mathbb{R}^{n \times p_2}$:

$$\mathrm{CKA}_{\mathrm{linear}}(\boldsymbol{X}, \boldsymbol{Y}) = \frac{\sum_{i=1}^{p_1} \sum_{j=1}^{p_2} \lambda_X^i \lambda_Y^j \langle \boldsymbol{u}_X^i, \boldsymbol{u}_Y^j \rangle^2}{\sqrt{\sum_{i=1}^{p_1} (\lambda_X^i)^2} \sqrt{\sum_{j=1}^{p_2} (\lambda_Y^j)^2}} \tag{2}$$

where $\boldsymbol{u}_X^i \in \mathbb{R}^n$ and $\boldsymbol{u}_Y^i \in \mathbb{R}^n$ are the ith normalized principal components of $\boldsymbol{X}$ and $\boldsymbol{Y}$, and λ_X^i and λ_Y^i are the amounts of variance that these principal components explain [19]. As the fraction of variance explained by the first principal component of each representation approaches 1, CKA becomes a measure of the squared cosine similarity between first principal components $\langle \boldsymbol{u}_X^1, \boldsymbol{u}_Y^1 \rangle^2$. Nguyen et al. [29] conclude that the block structure preserves and propagates a dominant first principal component across many hidden layers.

Datasets & Models: Our setup closely follows that of Nguyen et al. [29] and analyzes ResNets of varying depths and widths, trained on common image classification datasets CIFAR-10 and CIFAR-100 [20], as well as the medical imaging dataset Patch Camelyon [39]. These datasets are chosen to reflect the image statistics found in different domains, and all easily induce a block structure in reasonably sized ResNets.

The ResNet architecture design follows Zagoruyko and Komodakis [41], with the layers distributed evenly between three stages — each marked by a different feature map size — and the number of channels is doubled after each stage. To scale the model depth and width, we increase the number of layers and channels respectively. In experiments involving Shake-Shake regularization [7], the network is modified to have 3 branches that are combined in a stochastic fashion. More information on hyperparameters can be found in Appendix A.

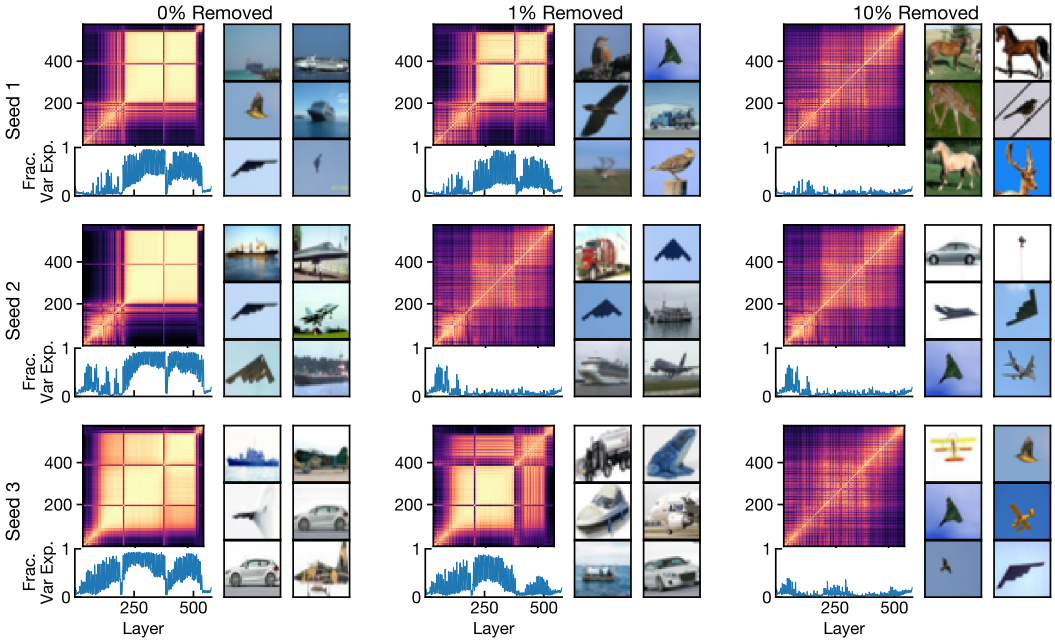

Figure 4: Removing dominant datapoints eliminates the block structure. Plots show the effect of removing examples with the largest projections on the first PC of layer 300 of ResNet-164 (1 ×) models. Columns reflect different numbers of examples removed; rows reflect models trained from different seeds. Within each group, the top left plot shows internal CKA heatmaps, the bottom left shows the fraction of variance explained by the first PC, and the images reflect the new examples with the largest projection on the first PC after data removal.

4 Dominant Datapoints

4.1 Datapoints that Activate the First PC

Motivated by previous evidence that the block structure propagates a dominant principal component across its constituent layers, we examine the distribution of the projections of each example's activations onto the first principal component. We find that the distribution is bimodal. Most examples have small projections, but some are orders of magnitude larger than the median (Figure 3).

We call these datapoints with large projections *dominant datapoints*, and find that they are consistent across the range of layers making up the block structure, as seen from each column of images in Figure 3 showing the dominant datapoints for two different layers in the same block structure. This explains why the first principal components of different layer activations inside the block structure are highly similar (Figure 2a), an observation made earlier in [29]. Moreover, this dominant datapoint phenomenon is present only in networks that also exhibit a block structure. As shown in Appendix Figure D.1, in networks without block structure, projections on the first principal component are unimodally distributed and the corresponding datapoints differ between layers.

Dominant datapoints are visually similar. In the left column of Figure 3, we observe that all datapoints have a blue background, although the precise shade of blue varies. However, the visual signals that the first principal component picks up on depend on the random seed used to train